

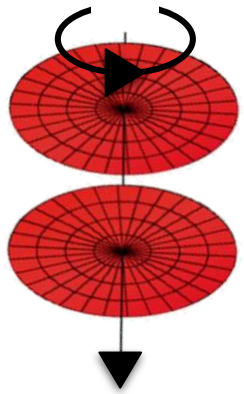
Theoretical studies of excitonic transitions in two dimensional materials under photoexcitation of scalar and vector vortex beams

1. Introduction : spin-orbit interaction (SOI)

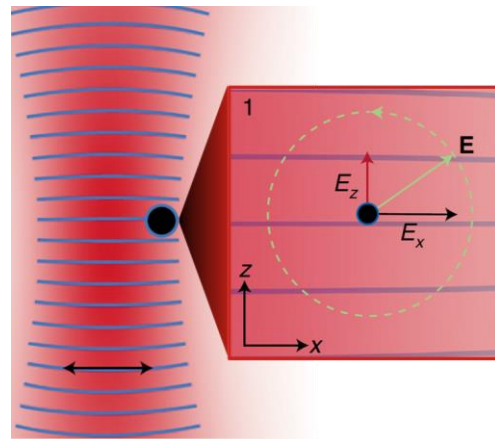
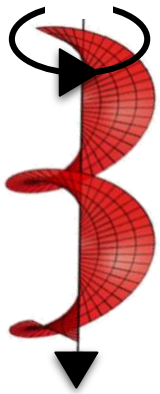
Scalar vortex beam : $|\sigma, \ell\rangle$

--- SOI in longitudinal component

Plane wave



$|1, 1\rangle$

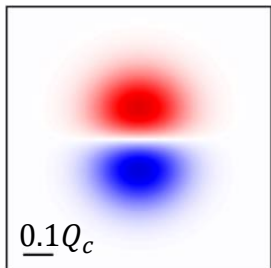


F. Cardano & L. Marrucci, *Nat. Photonics* 15, 72 (2021).

Re

q_y

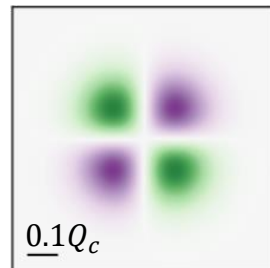
q_x



$e^{i\ell\phi_{q\parallel}}$



a.u.



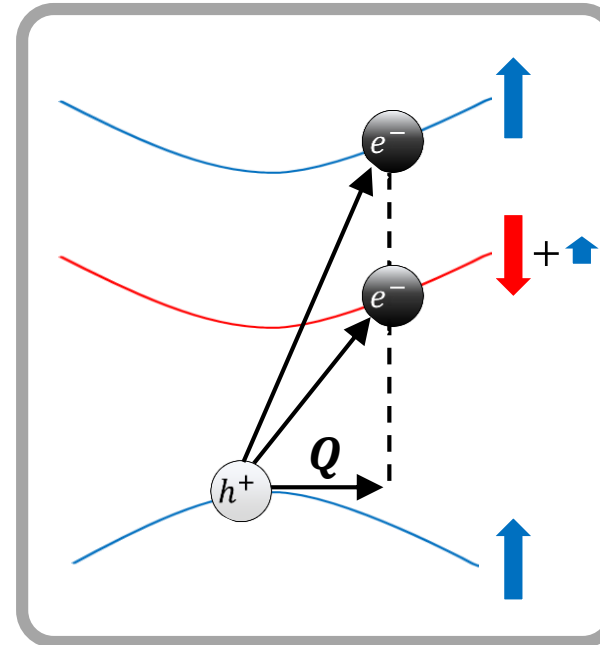
$e^{i(\sigma+\ell)\phi_{q\parallel}}$



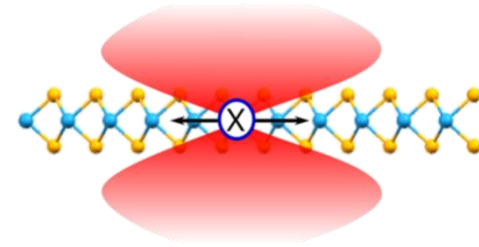
a.u.

Typical semiconductor

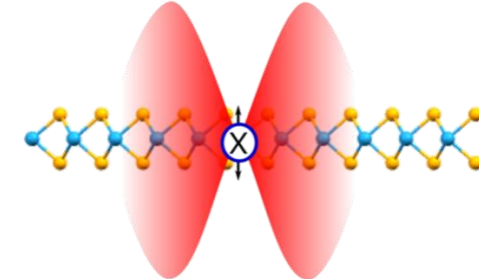
---SOI induced spin mixing



BX



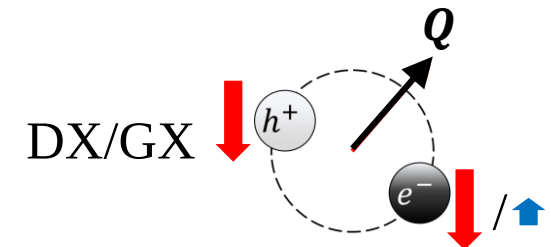
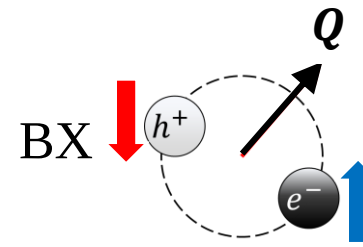
GX



DX



L. M. Schneider et al., *Sci. Rep.* 10, 8091 (2020)

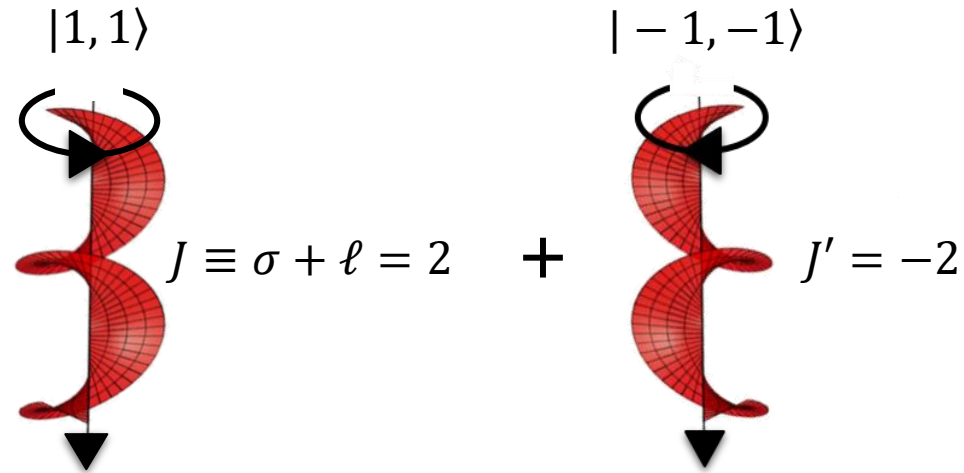


In plane wave vector : $q_{\parallel}$

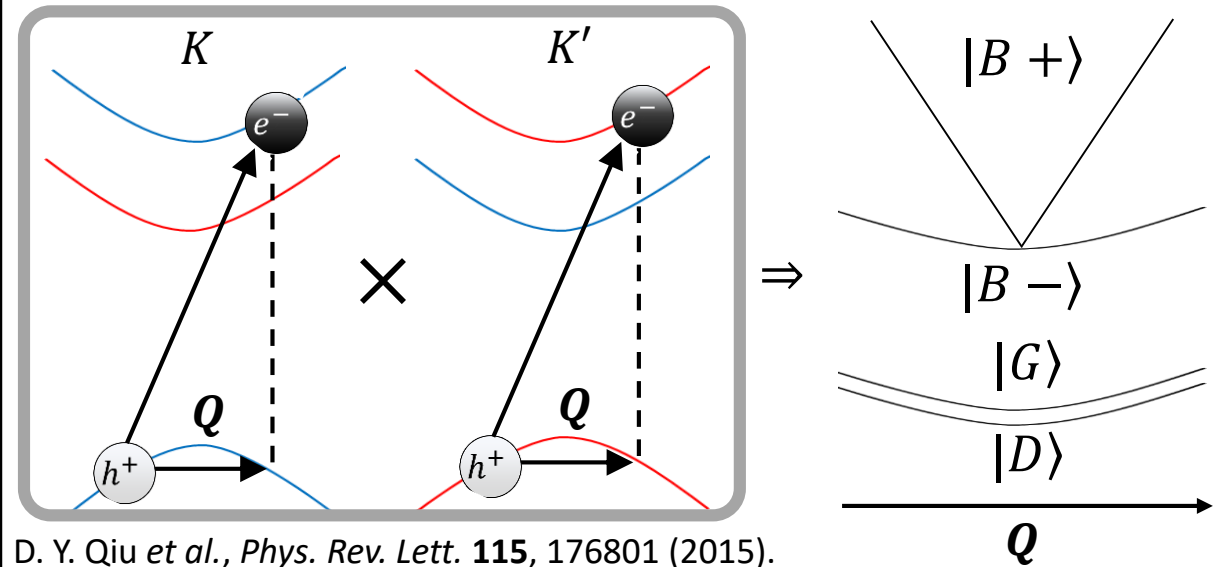
Center of mass momentum : Q

1. Introduction : mixed states

Vector vortex beam : $|\sigma, \ell, -\sigma, \ell'\rangle = |\sigma, \ell\rangle + |-\sigma, -\ell\rangle$



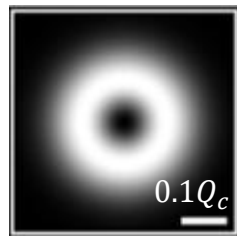
2D Transition metal dichalcogenide (TMD): valley-mixing
w/o e-h exchange w/ e-h exchange



D. Y. Qiu *et al.*, *Phys. Rev. Lett.* **115**, 176801 (2015).

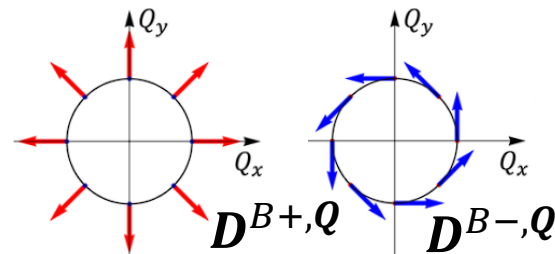
VVB

Transverse



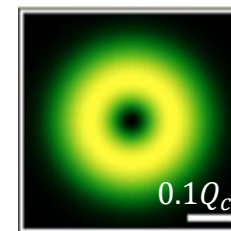
$\times$

Transition dipole moments



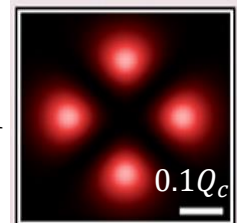
$\Rightarrow$

Total BX



O. J. G. Sanchez, G.-H. Peng *et al.*,
ACS Nano **18**, 11425 (2024)

Longitudinal

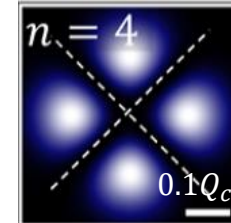


$\times$

$$D^{G,Q} = D^0 \hat{z}$$

$\Rightarrow$

GX



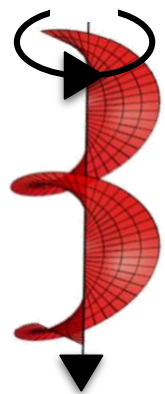
$$n = |\Delta J| = 2|\sigma + \ell|$$

Always even number

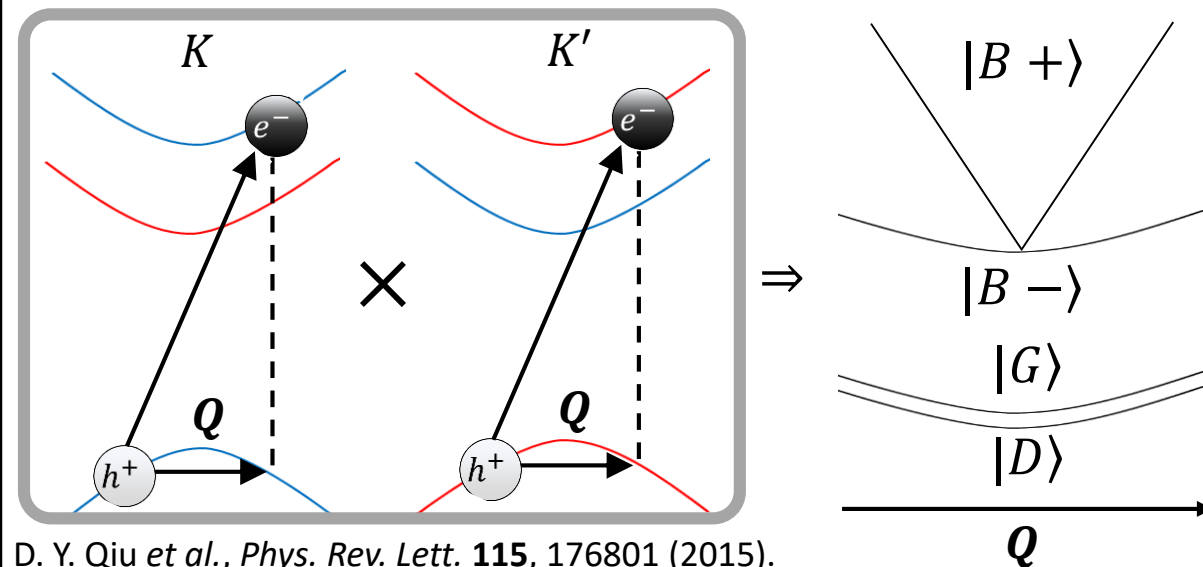
1. Introduction : mixed states

Scalar vortex beam : $|\sigma, \ell\rangle$

$$|\hat{\mathbf{e}} = \hat{\mathbf{x}}, 1\rangle$$

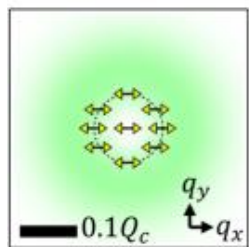


2D Transition metal dichalcogenide (TMD): valley-mixing
w/o e-h exchange w/ e-h exchange



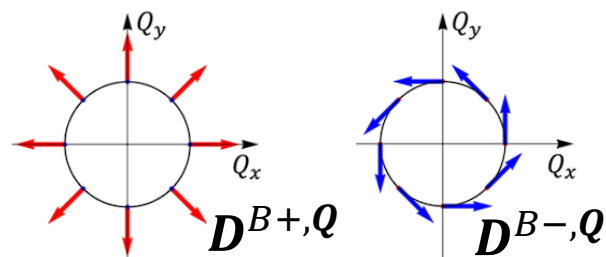
D. Y. Qiu et al., *Phys. Rev. Lett.* **115**, 176801 (2015).

G.-H. Peng et al., *Phys. Rev. B* 106, 155304 (2022).



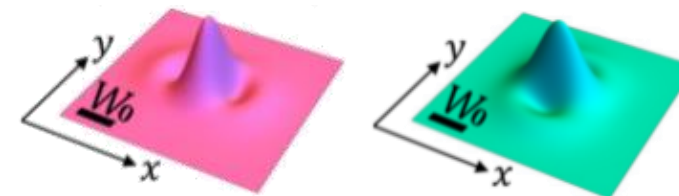
$$| |\hat{\mathbf{e}} = \hat{\mathbf{x}}, 1\rangle |^2$$

×



Transition dipole moments

⇒



Exciton wave packet

$$\propto e^{i\ell\phi_{q\parallel}}$$

$$|\Psi^{B\pm, \mathbf{Q}}\rangle = e^{i\delta(\mathbf{Q})} (e^{-i\phi_{\mathbf{Q}}} |\Psi^{K, \mathbf{Q}}\rangle \pm e^{i\phi_{\mathbf{Q}}} |\Psi^{K', \mathbf{Q}}\rangle) / \sqrt{2}$$

1. Introduction

2. Angular geometry in vector vortex beam

- $|\ell'| = |\ell|$: Transverse polarization and longitudinal SOI pattern
- $|\ell'| \neq |\ell|$: Robustness with angular momentum difference
- $|\ell'| \neq |\ell|$: Instantaneous field and intensity

3. Angular momentum difference governed optical transition

- Matched with winding number : Full-azimuthal excitation
- Mismatched with winding number : Directional excitation

4. Exciton wave packet

- Internal phase in exciton wave packet

Summary and outlook

1. Introduction

2. Angular geometry in vector vortex beam

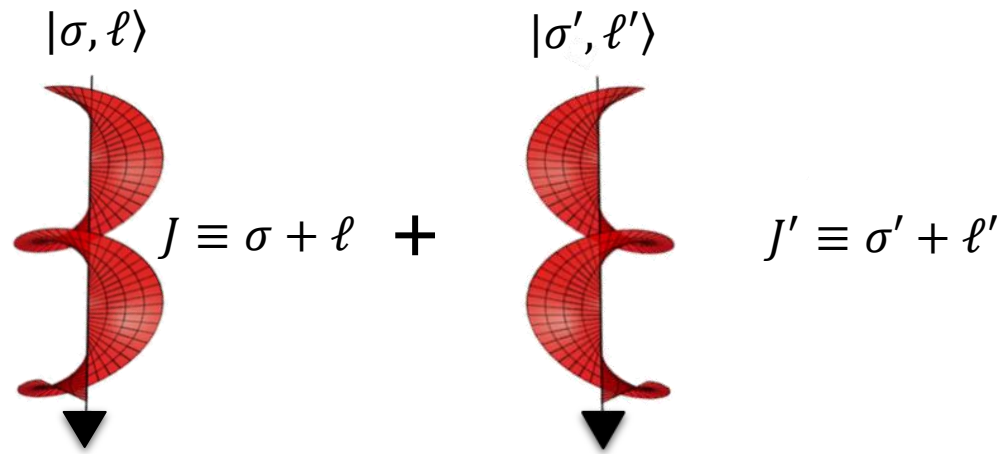
- $|\ell'| = |\ell|$: Transverse polarization and longitudinal SOI pattern
- $|\ell'| \neq |\ell|$: Robustness with angular momentum difference
- $|\ell'| \neq |\ell|$: Instantaneous field and intensity

3. Angular momentum difference governed optical transition

4. Exciton wave packet

Summary and outlook

2. VVB with $|\ell'| = |\ell|$: Polarization and SOI pattern



Vector vortex beam : $|\sigma, \ell, -\sigma, \ell'\rangle = |\sigma, \ell\rangle + |\sigma', \ell'\rangle$

Transverse component

$$\propto e^{i\bar{\ell}\phi_{q_{\parallel}}} \left\{ \cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}}\right) \hat{x} + \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}}\right) \hat{y} \right\}$$

Longitudinal component

$$\propto e^{i\bar{J}\phi_{q_{\parallel}}} \left\{ \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}}\right) \right\} \hat{z}$$

Average AM. $\{\bar{\ell} \equiv (\ell' + \ell)/2, \bar{J} \equiv (J' + J)/2\}$

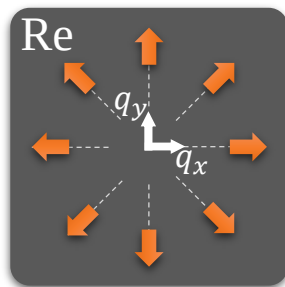
AM. difference $\{\Delta\ell \equiv \ell' - \ell, \Delta J \equiv J' - J\}$

$|\sigma, \ell, -\sigma, \ell'\rangle$

Transverse

Longitudinal

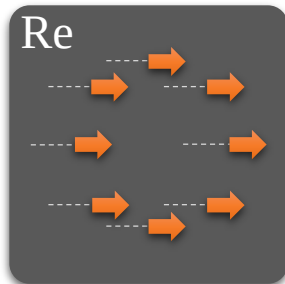
$|1, -1, -1, 1\rangle$



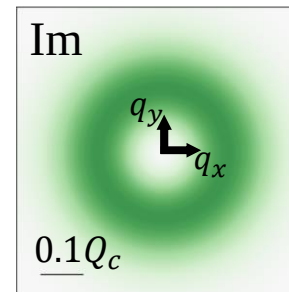
$$\Delta\ell = 2$$

linear polarization
(twisted when $\Delta\ell \neq 0$)

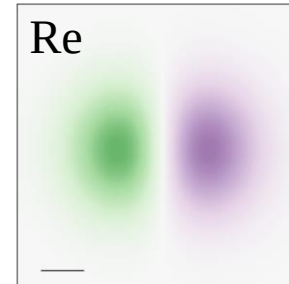
$|1, 0, -1, 0\rangle$



$$\Delta\ell = 0$$



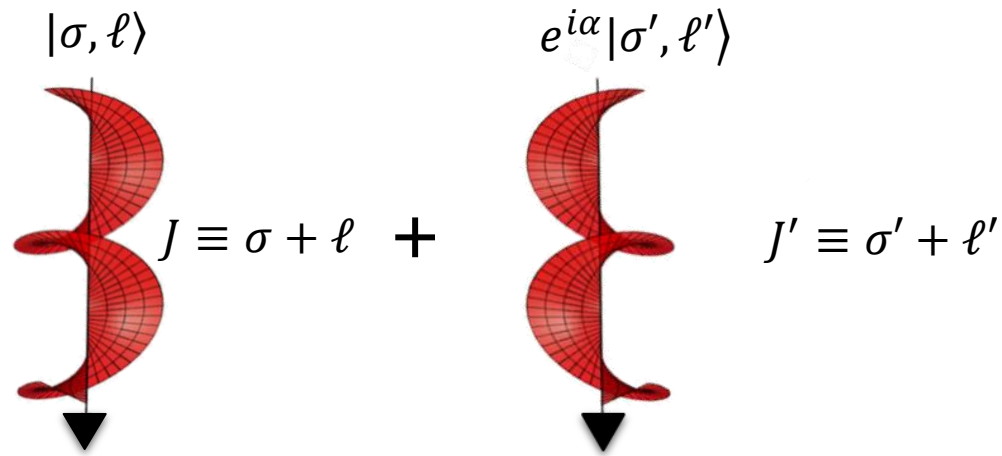
$$\Delta J = 0$$



$$\Delta J = -2$$

of node = $|\Delta J|$
(when absolutized)

2. VVB : Relative phase α



Vector vortex beam : $|\sigma, \ell, -\sigma, \ell'; \alpha\rangle = |\sigma, \ell\rangle + e^{i\alpha} |\sigma', \ell'\rangle$

Transverse component

$$\propto e^{i\bar{\ell}\phi_{q_{\parallel}}} \left\{ \cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{x} + \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{y} \right\}$$

Longitudinal component

$$\propto e^{i\bar{J}\phi_{q_{\parallel}}} \left\{ \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \right\} \hat{z}$$

Average AM. $\{\bar{\ell} \equiv (\ell' + \ell)/2, \bar{J} \equiv (J' + J)/2\}$

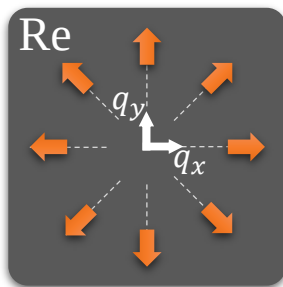
AM. difference $\{\Delta\ell \equiv \ell' - \ell, \Delta J \equiv J' - J\}$

$|\sigma, \ell, -\sigma, \ell'; \alpha\rangle$

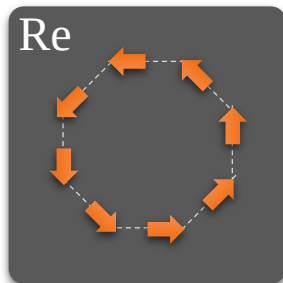
Transverse

Longitudinal

$|1, -1, -1, 1; \alpha\rangle$

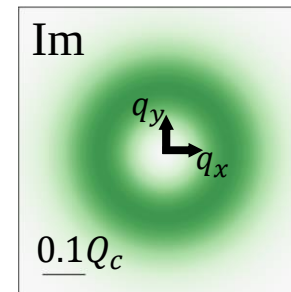


$\alpha = 0$

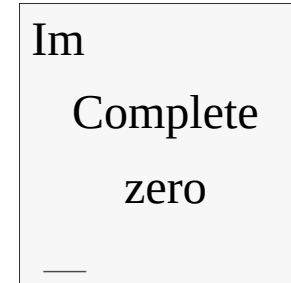


$\alpha = \pi$

Change :
Orientation



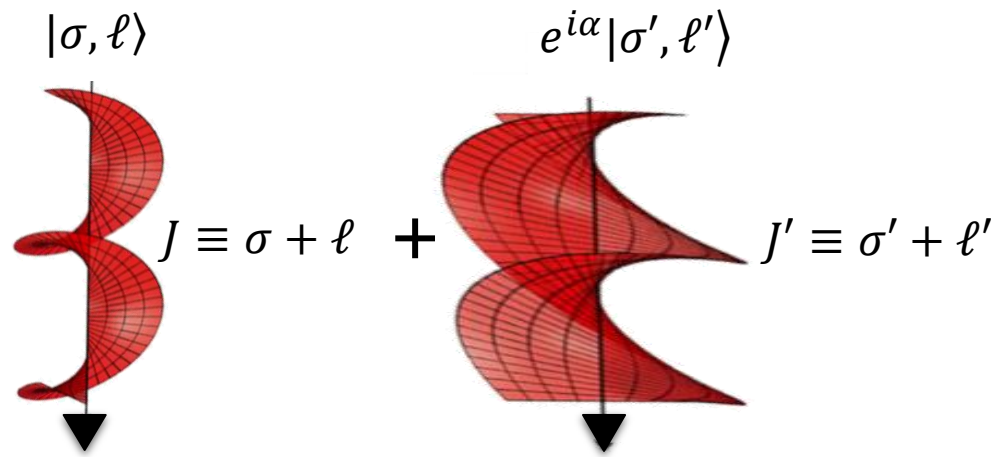
$\alpha = 0$



$\alpha = \pi$

Change :
Overall value

2. VVB with ($|\ell'| \neq |\ell|$): Robustness with angular momentum difference



Vector vortex beam : $|\sigma, \ell, -\sigma, \ell'; \alpha\rangle = |\sigma, \ell\rangle + e^{i\alpha}|\sigma', \ell'\rangle$

Transverse component

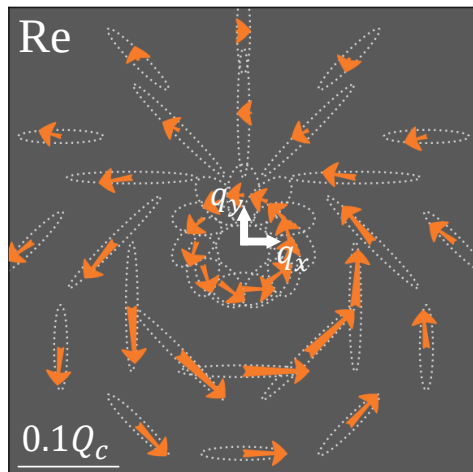
$$\propto e^{i\bar{\ell}\phi_{q_{\parallel}}} \left\{ \left[\cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) \hat{x} + \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) \hat{y} \right] G^+ + i[\dots]G^- \right\}$$

Longitudinal component

$$\propto e^{i\bar{J}\phi_{q_{\parallel}}} \left\{ \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) G^+ + i\sin\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) G^- \right\} \hat{z}$$

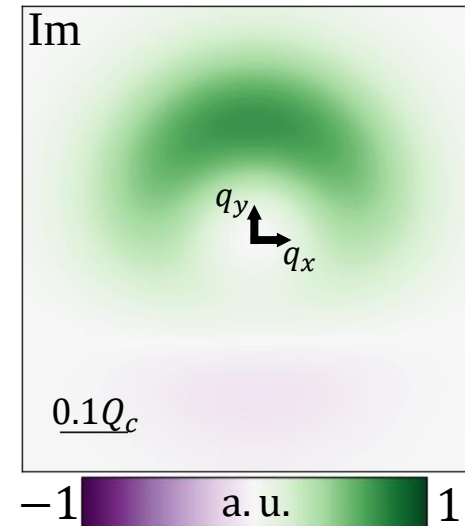
AM. relation $\{\bar{\ell} \equiv (\ell' + \ell)/2, \bar{J} \equiv (J' + J)/2, \Delta\ell \equiv \ell' - \ell, \Delta J \equiv J' - J\}$ Amplitude relation $\{G^{\pm} \equiv | |\sigma, \ell\rangle | \pm | |\sigma', \ell'\rangle |$

$ 1, -1, -1, 2; \alpha = 0\rangle$	Transverse	Longitudinal
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$\Delta\ell = 3$
polarization twist

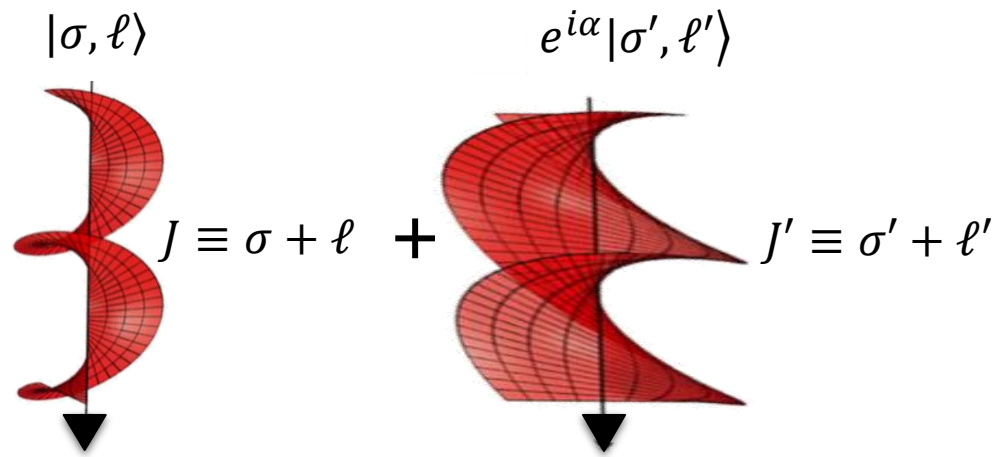
Subtle change:
Elliptical polarization



$|\Delta J| = 1$
one node (when absolutized)

Subtle change :
Counter-node SOI pattern

2. VVB with ($|\ell'| \neq |\ell|$): Estimate G^-



Vector vortex beam : $|\sigma, \ell, -\sigma, \ell'; \alpha\rangle = |\sigma, \ell\rangle + e^{i\alpha} |\sigma', \ell'\rangle$

Transverse component

$$\propto e^{i\bar{\ell}\phi_{q_{\parallel}}} \left\{ \left[\cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) \hat{x} + \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) \hat{y} \right] G^+ + i[\dots]G^- \right\}$$

Longitudinal component

$$\propto e^{i\bar{J}\phi_{q_{\parallel}}} \left\{ \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) G^+ + i\sin\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) G^- \right\} \hat{z}$$

AM. relation $\{\bar{\ell} \equiv (\ell' + \ell)/2, \bar{J} \equiv (J' + J)/2, \Delta\ell \equiv \ell' - \ell, \Delta J \equiv J' - J\}$ Amplitude relation $\{G^{\pm} \equiv | |\sigma, \ell\rangle | \pm | |\sigma', \ell'\rangle |$

$$||\sigma, \ell\rangle|^2$$

$$|1, -1, -1, 2; \alpha = 0\rangle$$

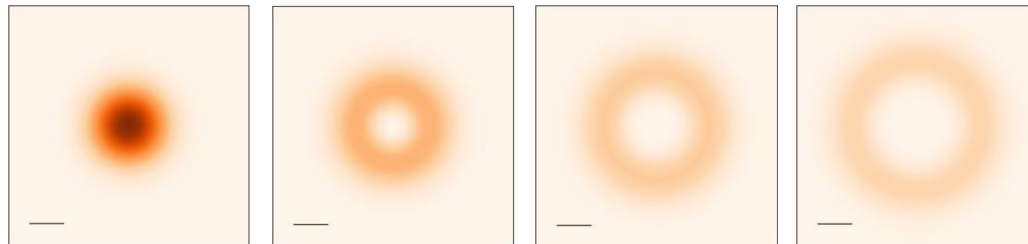
$\ell = 0$

$\ell = \pm 1$

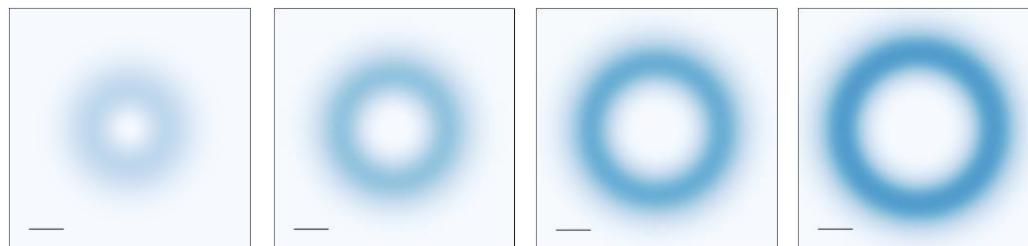
$\ell = \pm 2$

$\ell = \pm 3$

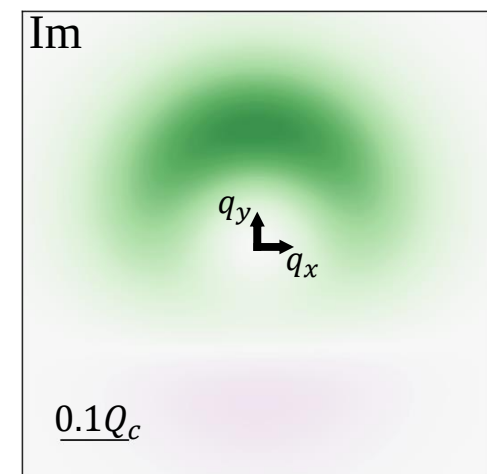
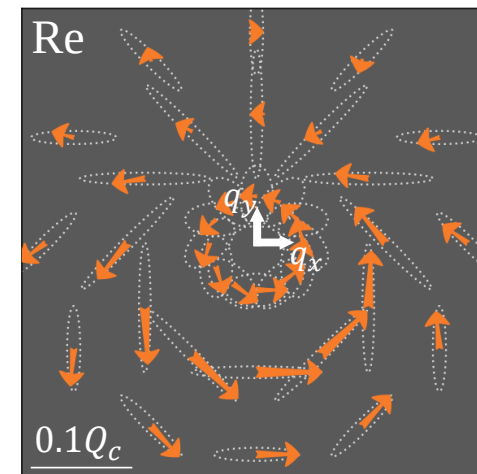
Transverse



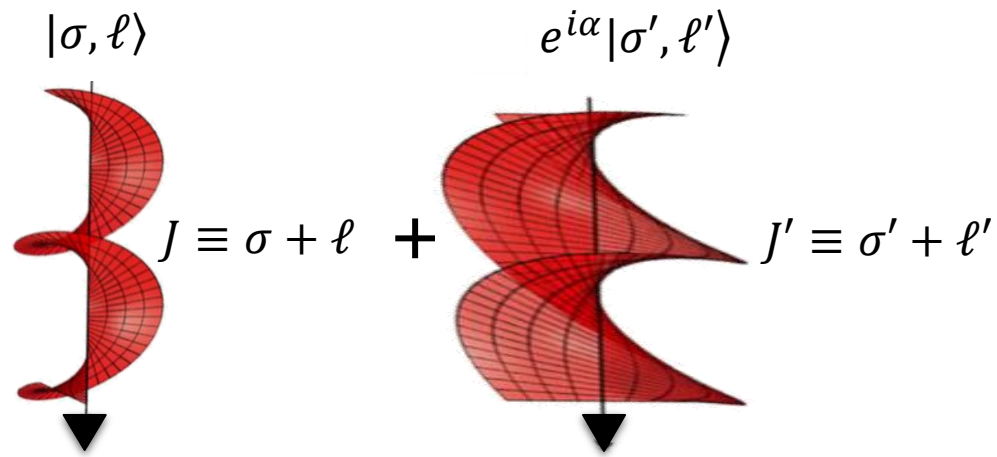
Longitudinal



Subtle change $\propto G^-$



2. VVB with ($|\ell'| \neq |\ell|$): Instantaneous field and intensity



Vector vortex beam : $|\sigma, \ell, -\sigma, \ell'; \alpha\rangle = |\sigma, \ell\rangle + e^{i\alpha}|\sigma', \ell'\rangle$

Transverse component

$$\propto e^{i\bar{\ell}\phi_{q_{\parallel}}} \left\{ \left[\cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) \hat{x} + \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) \hat{y} \right] G^+ + i[\dots]G^- \right\}$$

Longitudinal component

$$\propto e^{i\bar{J}\phi_{q_{\parallel}}} \left\{ \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) G^+ + i\sin\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) G^- \right\} \hat{z}$$

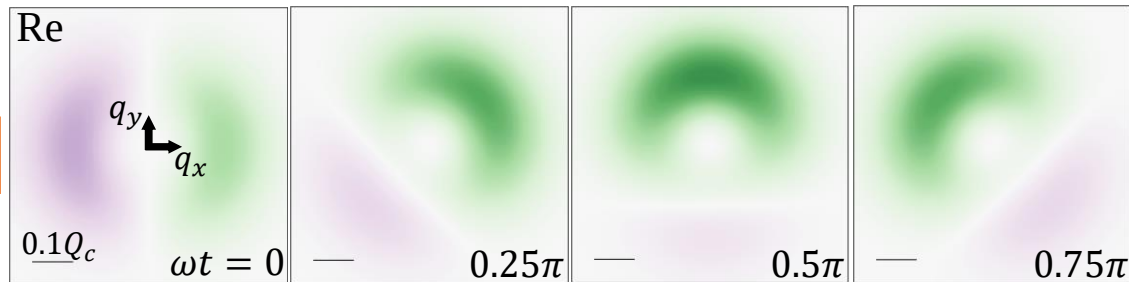
AM. relation $\{\bar{\ell} \equiv (\ell' + \ell)/2, \bar{J} \equiv (J' + J)/2, \Delta\ell \equiv \ell' - \ell, \Delta J \equiv J' - J$ Amplitude relation $\{G^{\pm} \equiv ||\sigma, \ell\rangle| \pm ||\sigma', \ell'\rangle|$

Longitudinal

$|1, -1, -1, 2; \alpha = 0\rangle$

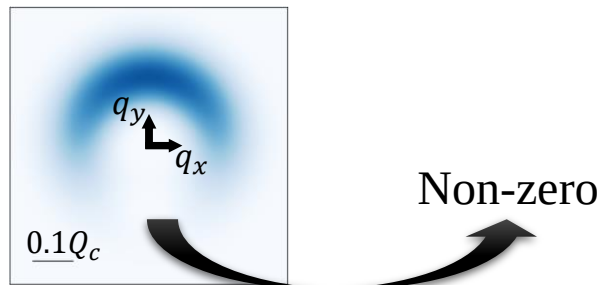
$|1, -1, -1, 2\rangle e^{-i\omega t}$

-1 a. u. 1

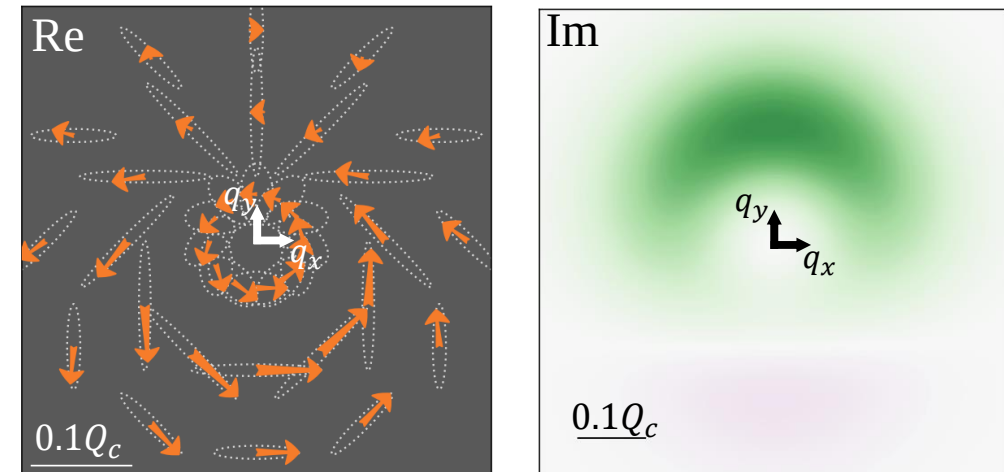


$||1, -1, -1, 2|^2$

0 a. u. 1



Subtle change $\propto G^-$



1. Introduction

2. Angular geometry in vector vortex beam

3. Angular momentum difference governed optical transition

- Matched with winding number : Full-azimuthal excitation
- Mismatched with winding number : Directional excitation

4. Exciton wave packet

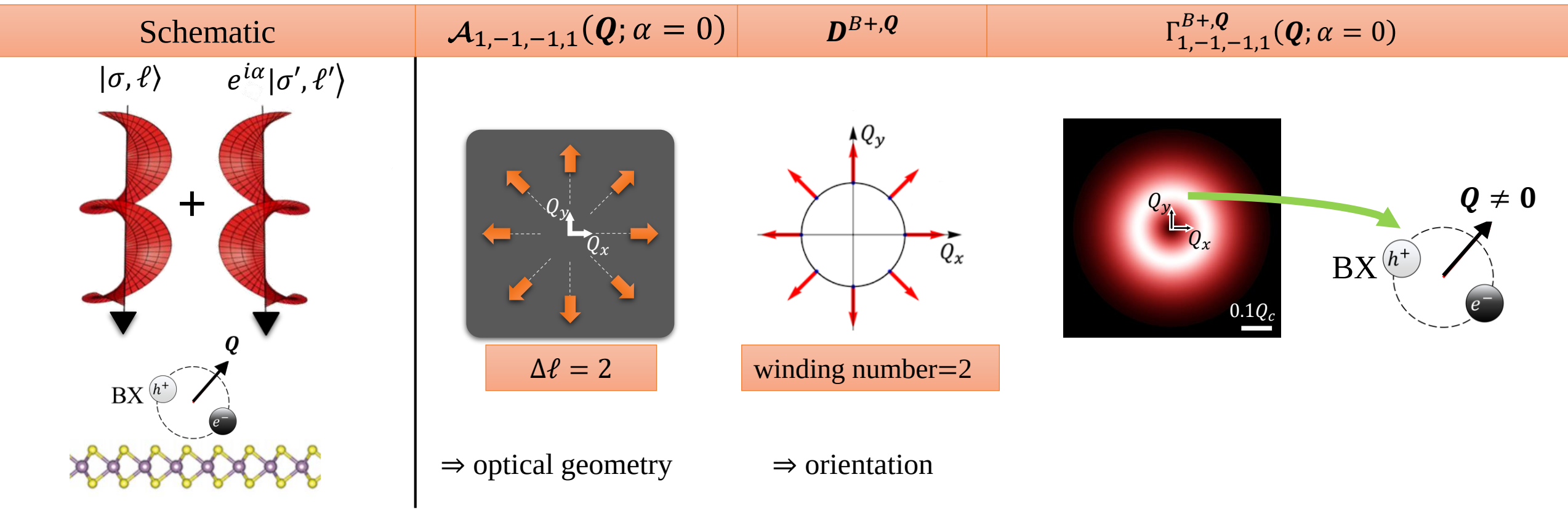
Summary and outlook

3. Transition rate

Following O. J. G. Sanchez, G.-H. Peng *et al.*, *ACS Nano* **18**, 11425 (2024), the transition rate is obtained as

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha) = \frac{2\pi}{\hbar} \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha) \right|^2 \rho(\hbar\omega = E_{S,Q}^X)$$

$$\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha) \propto \mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{Q}; \alpha) \cdot \mathbf{D}^{S,Q}$$



3. Matched with winding number : Full-azimuthal excitation

Following O. J. G. Sanchez, G.-H. Peng *et al.*, *ACS Nano* **18**, 11425 (2024), the transition rate is obtained as

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha) = \frac{2\pi}{\hbar} \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha) \right|^2 \rho(\hbar\omega = E_{S,Q}^X)$$

$$\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha) \propto \mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{Q}; \alpha) \cdot \mathbf{D}^{S,Q}$$

Vector vortex beam : $\mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{Q}; \alpha) = |\sigma, \ell, -\sigma, \ell'; \alpha\rangle$

Transverse

$$\propto e^{i\bar{\ell}\phi_{q\parallel}} \left\{ \cos\left(\frac{\Delta\ell}{2}\phi_{q\parallel} + \frac{\alpha}{2}\right) \hat{x} + \sin\left(\frac{\Delta\ell}{2}\phi_{q\parallel} + \frac{\alpha}{2}\right) \hat{y} \right\}$$

Longitudinal

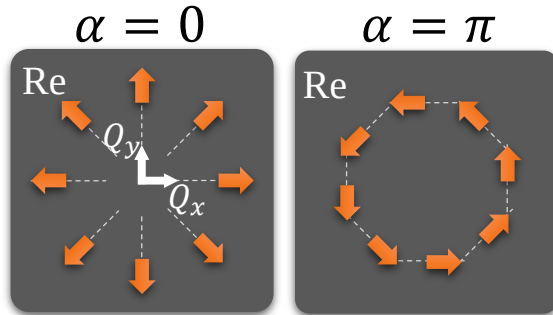
$$\propto e^{i\bar{J}\phi_{q\parallel}} \left\{ \cos\left(\frac{\Delta J}{2}\phi_{q\parallel} + \frac{\alpha}{2}\right) \right\} \hat{z}$$

$\mathcal{A}_{1,-1,-1,1}(\mathbf{Q}; \alpha)$

$\mathbf{D}^{S,Q}$

Transition rate

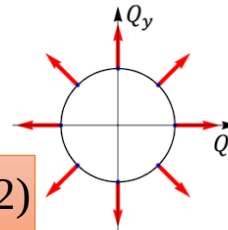
Transverse
($\Delta\ell = 2$)



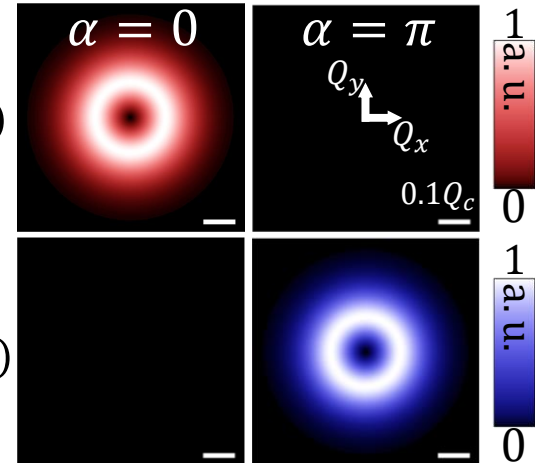
$\times$

$\mathbf{D}^{B+,Q}$

(winding number=2)

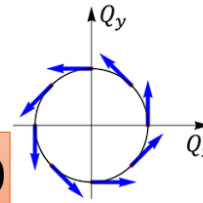


$$\Rightarrow \Gamma_{1,-1,-1,1}^{B+,Q}(\mathbf{Q}; \alpha)$$



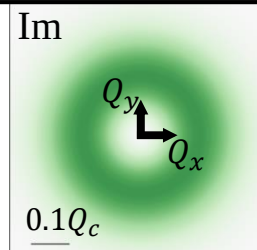
$\mathbf{D}^{B-,Q}$

(winding number=2)



$$\Rightarrow \Gamma_{1,-1,-1,1}^{B-,Q}(\mathbf{Q}; \alpha)$$

Longitudinal
($\Delta J = 0$)



Complete
zero

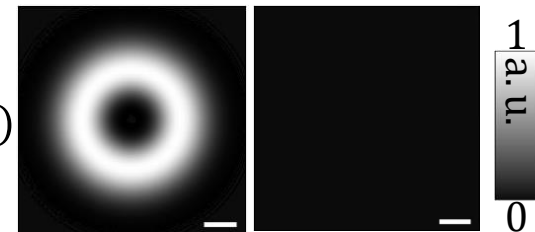
$\times$

$\mathbf{D}^{G,Q}$

(winding number $\Rightarrow 0$)

$\mathbf{D}^0 \hat{z}$

$$\Rightarrow \Gamma_{1,-1,-1,1}^{G,Q}(\mathbf{Q}; \alpha)$$



3. Mismatched with winding number : Directional excitation

Following O. J. G. Sanchez, G.-H. Peng *et al.*, *ACS Nano* **18**, 11425 (2024), the transition rate is obtained as

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha) = \frac{2\pi}{\hbar} \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha) \right|^2 \rho(\hbar\omega = E_{S,Q}^X)$$

$$\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha) \propto \mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{Q}; \alpha) \cdot \mathbf{D}^{S,Q}$$

Vector vortex beam : $\mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{Q}; \alpha) = |\sigma, \ell, -\sigma, \ell'; \alpha\rangle$

Transverse

$$\propto e^{i\bar{\ell}\phi_{q_{\parallel}}} \left\{ \left[\cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) \hat{x} + \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) \hat{y} \right] G^+ + i[\dots] G^- \right\}$$

Longitudinal

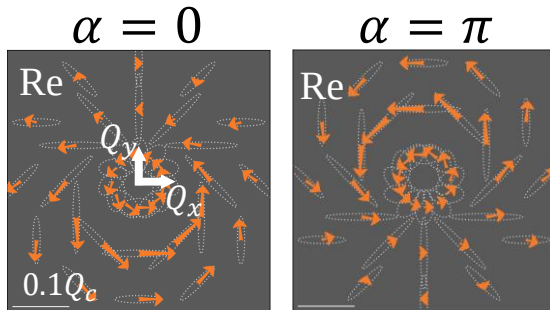
$$\propto e^{i\bar{J}\phi_{q_{\parallel}}} \left\{ \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) G^+ + i \sin\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \alpha_{\text{const.}}\right) G^- \right\} \hat{z}$$

$\mathcal{A}_{1,-1,-1,2}(\mathbf{Q}; \alpha)$

$\mathbf{D}^{S,Q}$

Transition rate

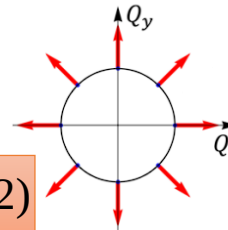
Transverse
($\Delta\ell = 3$)



$\times$

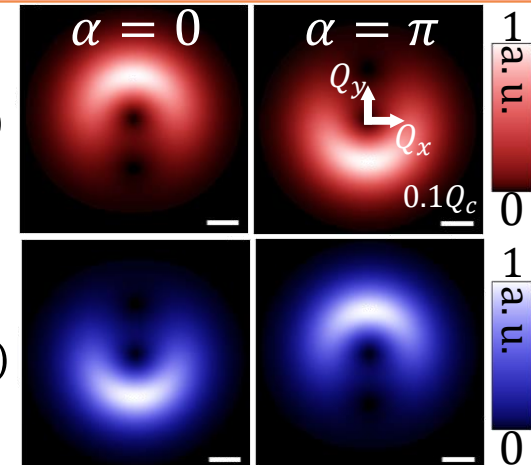
$\mathbf{D}^{B+,Q}$

(winding number=2)



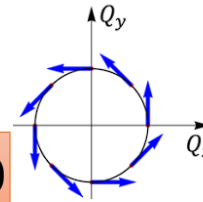
$\Rightarrow$

$\Gamma_{1,-1,-1,2}^{B+,Q}(\mathbf{Q}; \alpha)$



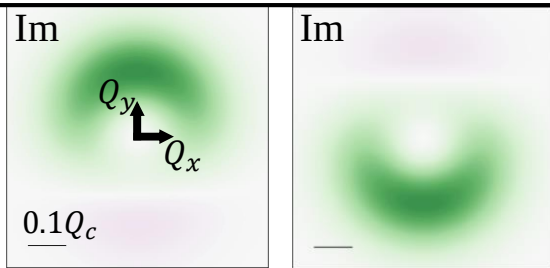
$\mathbf{D}^{B-,Q}$

(winding number=2)



$\Gamma_{1,-1,-1,2}^{B-,Q}(\mathbf{Q}; \alpha)$

Longitudinal
($\Delta J = 1$)



$\times$

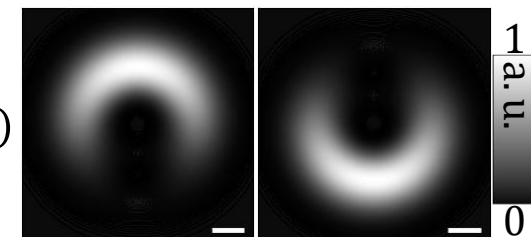
$\mathbf{D}^{G,Q}$

(winding number=0)

$\mathbf{D}^0 \hat{z}$

$\Rightarrow$

$\Gamma_{1,-1,-1,2}^{G,Q}(\mathbf{Q}; \alpha)$



1. Introduction
2. Angular geometry in vector vortex beam
3. Angular momentum difference governed optical transition
4. Exciton wave packet
 - Internal phase in exciton wave packet

Summary and outlook

4. Global phase problem

Superposition of finite-momentum exciton state :

$$|\Psi(t)\rangle = |GS\rangle + \sum_S \sum_Q \tilde{c}^{S,Q(1)}(t) e^{-i\omega^{S,Q}t} |\Psi^{S,Q}\rangle$$

exciton wave packet :

$$f_{\hat{\mathbf{e}},\ell}^{S,Q}(\mathbf{R}_c, t) \equiv \frac{1}{\sqrt{\Omega}} \sum_Q \tilde{c}^{S,Q(1)}(t) e^{i\mathbf{Q}\cdot\mathbf{R}_c}$$

G.-H. Peng *et al.*, *Phys. Rev. B* 106, 155304 (2022). Ω : area of 2D TMD

1. Global phase

$$|\Psi^{B\pm,Q}\rangle = e^{i\delta(Q)} (e^{-i\phi_Q} |\Psi^{K,Q}\rangle \pm e^{i\phi_Q} |\Psi^{K',Q}\rangle) / \sqrt{2}$$

2. Redefine the exciton wave packet

$$\Phi_{\hat{\mathbf{e}},\ell}^{S,Q}(\mathbf{R}_c, \mathbf{r}, t) \equiv \frac{1}{\sqrt{\Omega}} \sum_Q \tilde{c}^{S,Q(1)}(t) \Psi^{S,Q}(\mathbf{R}_c, \mathbf{r})$$

3. Analytically evaluate $\Psi^{S,Q}(\mathbf{R}_c, \mathbf{r})$

- Exciton wave function :

$$\Psi^{S,Q}(\mathbf{r}_e, \mathbf{r}_h) = \sum_{vck} \Lambda^{S,Q}(vck) \psi_{c,k+Q}(\mathbf{r}_e) \psi_{c,k}^*(\mathbf{r}_h)$$

- Bloch theory ($\mathbf{R}_c = a\mathbf{r}_e + b\mathbf{r}_h = \mathbf{r}_e, \mathbf{r} = \mathbf{r}_e - \mathbf{r}_h$):

$$\Psi^{S,Q}(\mathbf{R}_c, \mathbf{r})$$

$$= e^{i\mathbf{Q}\cdot\mathbf{R}_c} \sum_{vck} e^{ik\cdot\mathbf{r}} \Lambda^{S,Q}(vck) u_{c,k+Q}(\mathbf{R}_c) u_{c,k}^*(\mathbf{R}_c - \mathbf{r})$$

Zeroth-order in Taylor expansion

$$\approx \underbrace{e^{i\mathbf{Q}\cdot\mathbf{R}_c}}_{\text{Plane wave}} \underbrace{\sum_{vck} e^{ik\cdot\mathbf{r}} \Lambda^{S,0}(vck) u_{c,k}(\mathbf{R}_c) u_{c,k}^*(\mathbf{R}_c - \mathbf{r})}_{\text{Internal structure } U^{S,0}(\mathbf{R}_c, \mathbf{r})}$$

Plane wave

Internal structure $U^{S,0}(\mathbf{R}_c, \mathbf{r})$

- Wave packet is estimated with $\tilde{c}^{S,Q(1)}(t) \approx \frac{\tilde{\gamma}^{0(t)}}{\hbar\omega^0} \tilde{M}_{\hat{\mathbf{e}},\ell}^{S,Q}$,

$$\Phi_{\hat{\mathbf{e}},\ell}^{S,Q}(\mathbf{R}_c, \mathbf{r}) = \frac{1}{\hbar\omega^0\sqrt{\Omega}} U^{S,0}(\mathbf{R}_c, \mathbf{r}) \sum_Q \tilde{M}_{\hat{\mathbf{e}},\ell}^{S,Q} e^{i\mathbf{Q}\cdot\mathbf{R}_c}$$

4. Valley exciton wave packet

Redefine and rewrite with $\Psi^{S,Q}(\mathbf{R}_c, \mathbf{r}) \approx e^{i\mathbf{Q} \cdot \mathbf{R}_c} U^{S,0}(\mathbf{R}_c, \mathbf{r})$

$$\Phi_{\hat{\mathbf{e}},\ell}^{K,Q}(\mathbf{R}_c, \mathbf{r}) = \frac{1}{\hbar\omega^0\sqrt{\Omega}} U^{S,0}(\mathbf{R}_c, \mathbf{r}) \sum_Q \tilde{M}_{\hat{\mathbf{e}},\ell}^{S,Q} e^{i\mathbf{Q} \cdot \mathbf{R}_c}$$

$$\tilde{M}_{\hat{\mathbf{e}},\ell}^{S,Q} \propto \mathcal{A}_{\hat{\mathbf{e}},\ell}(\mathbf{Q}) \cdot \mathbf{D}^{S,Q}$$

$$(\mathbf{R}_c = a\mathbf{r}_e + b\mathbf{r}_h = \mathbf{r}_e, \mathbf{r} = \mathbf{r}_e - \mathbf{r}_h)$$

Phase from $U^{S,0}(\mathbf{R}_c, \mathbf{r})$ is neglected

$$U^{S,0}(\mathbf{R}_c, \mathbf{r}) = \sum_{vck} e^{i\mathbf{k} \cdot \mathbf{r}} \Lambda^{S,0}(vck) u_{c,k}(\mathbf{R}_c) u_{c,k}^*(\mathbf{R}_c - \mathbf{r})$$

1. Recombination

$$\mathbf{r} = \mathbf{0}$$

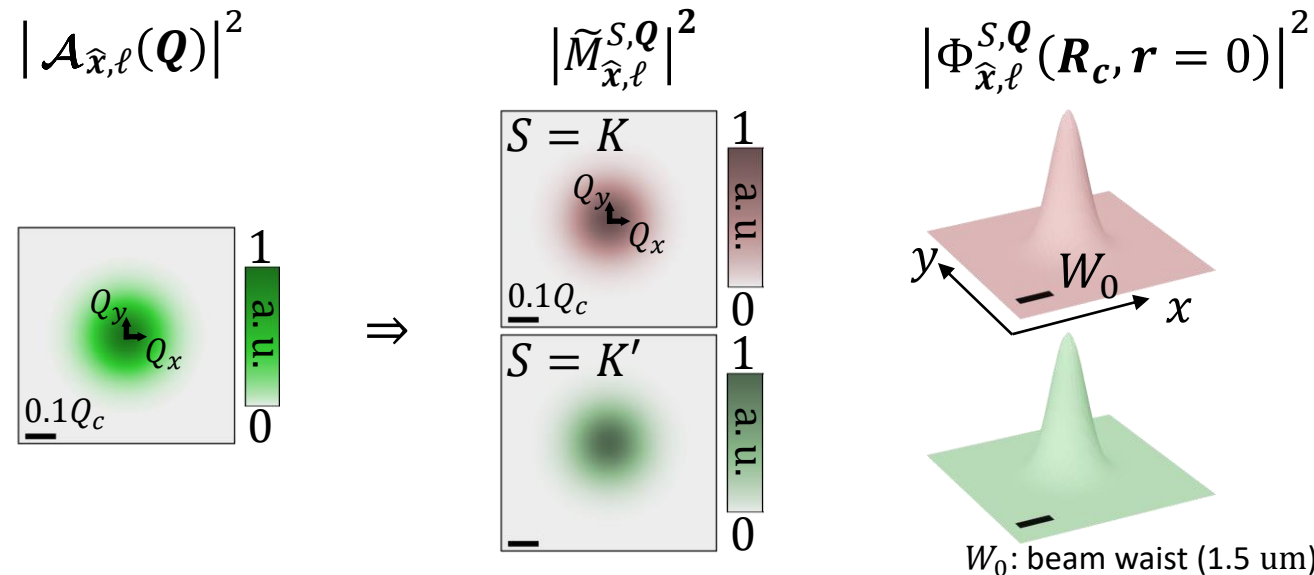
2. 2D hydrogen model

$$\Lambda^{1s,0}(vck) = \frac{\sqrt{2\pi}a_B^*}{\left[1 + \left(\frac{a_B^*k}{2}\right)^2\right]^{3/2}}$$

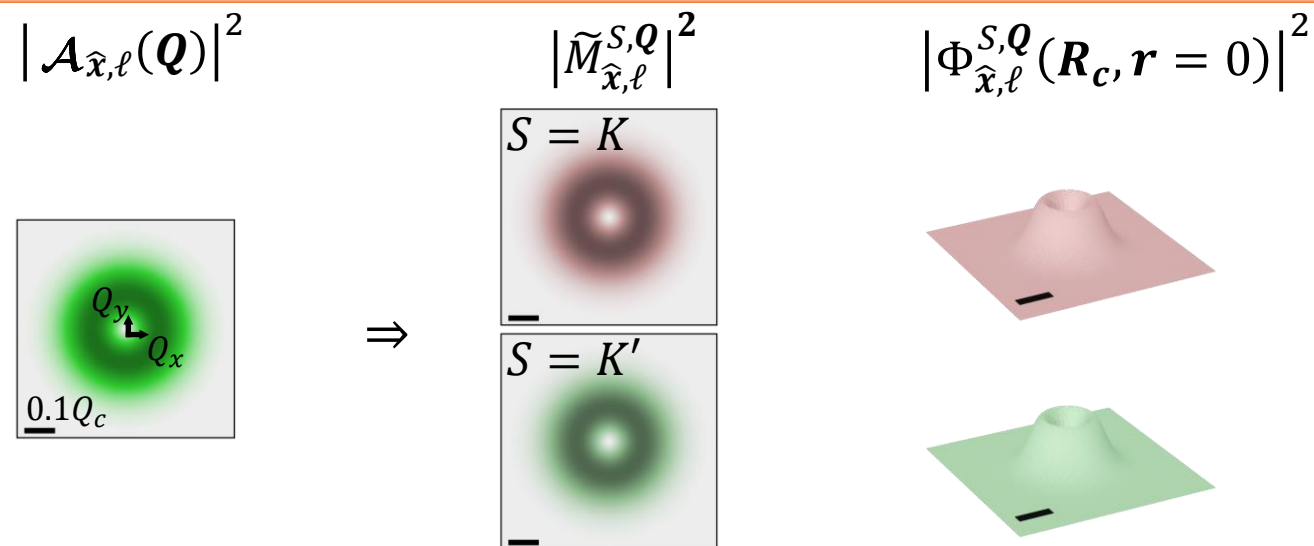
3. On lattice site

$$u_{n,k}(\mathbf{r}') = u_{n,k}(\mathbf{r}' + \mathbf{R})$$

Valley exciton wave packets : $\ell = 0$



Valley exciton wave packets : $\ell = \pm 1$



4. Internal phase in exciton wave packet

Valley exciton

$$\Phi_{\hat{\varepsilon},\ell}^{K,Q}(\mathbf{R}_c, \mathbf{r}) = \frac{1}{\hbar\omega^0\sqrt{\Omega}} |U^{K/K',0}(\mathbf{R}_c, \mathbf{r})|$$

$$\times \boxed{e^{i\text{Arg}[U^{K,0}(\mathbf{R}_c, \mathbf{r})]}} \sum_{\mathbf{Q}} \tilde{M}_{\hat{\varepsilon},\ell}^{K/K',Q} e^{i\mathbf{Q}\cdot\mathbf{R}_c}$$

$\approx$ fixed phase

Phase from $U^{S,0}(\mathbf{R}_c, \mathbf{r})$ is neglected

$$U^{S,0}(\mathbf{R}_c, \mathbf{r}) \approx \sum_{vck} e^{ik\cdot\mathbf{r}} \Lambda^{S,0}(vck) u_{c,k}(\mathbf{R}_c) u_{c,k}^*(\mathbf{R}_c - \mathbf{r})$$

1. Recombination

$$\mathbf{r} = \mathbf{0}$$

2. 2D hydrogen model

$$\Lambda^{1s,0}(vck) = \frac{\sqrt{2\pi}a_B^*}{\left[1 + \left(\frac{a_B^*k}{2}\right)^2\right]^{3/2}}$$

3. On lattice site

$$u_{n,k}(\mathbf{r}') = u_{n,k}(\mathbf{r}' + \mathbf{R})$$

Valley-mixed exciton

$$\Phi_{\hat{\varepsilon},\ell}^{B+/B-,Q}(\mathbf{R}_c, \mathbf{r}) = \frac{1}{\hbar\omega^0\sqrt{2\Omega}} |U^{K,0}(\mathbf{R}_c, \mathbf{r})|$$

$$\times \left[\boxed{e^{i\text{Arg}[U^{K,0}(\mathbf{R}_c, \mathbf{r})]}} \sum_{\mathbf{Q}} \left(e^{-i\ell\phi_Q} \tilde{M}_{\hat{\varepsilon},\ell}^{K,Q} \pm e^{i\ell\phi_Q} \tilde{M}_{\hat{\varepsilon},\ell}^{K',Q} \right) e^{i\mathbf{Q}\cdot\mathbf{R}_c} e^{-i\ell\phi_Q} \right.$$

$$\left. \pm \boxed{e^{-i\text{Arg}[U^{K,0}(\mathbf{R}_c, \mathbf{r})]}} \sum_{\mathbf{Q}} \left(e^{-i\ell\phi_Q} \tilde{M}_{\hat{\varepsilon},\ell}^{K,Q} \pm e^{i\ell\phi_Q} \tilde{M}_{\hat{\varepsilon},\ell}^{K',Q} \right) e^{i\mathbf{Q}\cdot\mathbf{R}_c} e^{i\ell\phi_Q} \right]$$

relative phase
contribution

Further numerical calculation is needed

Summary and outlook

Summary :

- The angular geometry of VVBs exists in **both the longitudinal and transverse** components **beyond the equal-OAM condition** ($|\ell| = |\ell'|$) . It is **preserved in both the instantaneous field and the intensity** distribution, provided that the constituent beams maintain sufficient spatial overlap.
- The TAM/OAM difference is the key parameter governing the angular geometry. Its matching or mismatching with the exciton winding number corresponding to **optical geometry and exciton transition dipole moments orientation**, which enables theoretically **full-azimuthal and directional excitation**.
- For valley-mixed exciton wave packets, the **phase of the exciton wave function** must be considered together with the optical phase to accurately determine the wave-packet structure.

Outlook

- Selective excitation of excitons: Use tailored vector vortex beams to selectively drive valley-mixed bright and gray excitons within specific center-of-mass momentum ranges.
- Topology of light–exciton coupling: How the topological structures of Laguerre-Gaussian beams are transferred to exciton wave packets .